

# Access to Interpretation: How Formal Cues Ground Interpretive Alt Text for Paintings

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## Abstract

Alt text is a core accessibility feature for blind and low vision (BLV) users, yet prevailing conventions emphasize objective description. While effective for many functional images, these norms can under-serve paintings, where meaning emerges through formal attributes and interpretation. We qualitatively analyze alt text for 91 paintings from an open-access collection and find heavy reliance on templated descriptions that are often redundant and provide uneven support for interpretive engagement. We contribute a provisional taxonomy that links objective formal cues to plausible interpretive attributes that distinguishes observation from interpretation. We position this taxonomy as a scaffold for supporting accountable interpretive access in painting alt text.

## CCS Concepts

• Human-centered computing → Accessibility design and evaluation methods.

## Keywords

interpretive alt text, creativity support, accessibility, art analysis

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## 1 Introduction

Alt text provides blind and low vision (BLV) screen reader users with informational context about an image, typically through concise, objective descriptions [20]. Traditionally, alt text prioritizes conveying information that is essential and task-relevant [12]. Paintings, however, are interpretive media pieces where meaning is often derived from formal attributes such as composition, color, lighting, and style [9]. This often leaves attributes such as inference, mood, and symbolism outside the scope of what alt text should provide. When alt text practices are applied to paintings, this perspective can reduce alt text to surface facts or catalog metadata, leaving

readers without support for the interpretive work that paintings are meant to evoke.

In this work, we approach the following research questions:

**RQ1:** How are alt text practices currently represented in the context of paintings?

**RQ2:** How can interpretive content be systematically derived from observable formal cues in paintings?

Through insights gathered from a close reading of open-access paintings, we contribute a provisional taxonomy of interpretive response that maps observable formal cues (i.e., composition, lighting, gesture, style) to interpretive responses (i.e., affect, narrative, symbolism). This taxonomy reframes subjectivity as a structured, accountable translation from observation, rather than a departure from objectivity. Through this work, we offer a scaffold for designing and evaluating interpretive alt text for paintings.

## 2 Related Work

Alt text is commonly framed as an accessibility practice oriented toward communicating an image’s function and essential content within the surrounding webpage context, with guidance emphasizing concise, purpose-driven descriptions that support screen reader access [1, 6, 19]. Practical guidance for writing alt text similarly foregrounds brevity and relevance, often privileging what can be straightforwardly stated as present in the image [7]. Within this framing, high-quality alt text is frequently characterized as objective and efficient, reflecting assumptions that neutrality and brevity best support screen reader access.

Prior HCI research has complicated this “objective, concise” framing by showing that alt text quality is shaped by social context [12], author perspective [12], and the tools and workflows that scaffold authorship [14]. Rather than a neutral transcription, alt text production involves decisions about salience, ordering, and omission that can meaningfully affect what BLV people can access [16]. Prior work highlights how alt text is negotiated within broader sociotechnical systems, including platform expectations, coordination among stakeholders, and the tensions between compliance-oriented practices and expressive or context-rich descriptions [3, 12]. Adjacent work on image description more broadly has further highlighted how representational choices in language affect downstream comprehension, while also demonstrating how standardized description structures can flatten nuance or obscure interpretive relationships between visual features and meaning [11].

A key reason paintings conflict with traditional alt text conventions is that artworks are designed to invite interpretation rather than only communicate propositional facts. Foundational accounts



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of aesthetic experience emphasize perception, meaning-making, and the role of the viewer in constructing significance through interaction with form and context [10]. Disability studies and first-person writing on visual culture further underscore that interpretation is not an optional “extra,” but part of how access, expertise, and authority are negotiated in encounters with art [13]. Recent accessibility research identifies an “interpretation gap” in accessible art engagement, showing that descriptive accuracy alone may not provide sufficient scaffolding for how artworks are understood, contextualized, or felt [8]. For paintings specifically, meaning is often carried through formal attributes such as composition, light, color, and style, as well as through culturally situated iconography, requiring translations from observable cues to interpretive claims [4].

In response to these challenges, creative and critical approaches to alt text have emerged that treat description as a site of voice, aesthetics, and relational engagement, rather than solely as a mechanism for informational adequacy [2]. Related work in interpretive media contexts similarly argues that informative and interpretive goals are not mutually exclusive, but represent design tensions that can be productively negotiated through structure, stance, and transparency [21]. Rather than attempting to eliminate interpretation, these perspectives call for making interpretive processes explicit and accountable, allowing readers to understand how meaning is being constructed from visual features [8, 12, 14]. Building on this body of work that positions alt text as a form of relational engagement, we investigate how to make interpretive processes more explicit and accountable, informing future HCI efforts for expanding interpretive alt text provision.

### 3 Data Collection and Analysis

We scraped painting records from the Art Institute of Chicago’s Open Access Image dataset and assembled a corpus of the first 91 out of 1980 paintings for analysis. We restricted inclusion to entries labeled as painting, open access, and having alt text. This filtering ensured the dataset reflected publicly accessible images with institution-provided alt text suitable for qualitative analysis. At the time of data collection, the Art Institute of Chicago did not publicly document detailed authoring guidelines specific to alt text in its online gallery. While the institution provides audio guides and accessibility resources for select works, we were unable to identify publicly available policies outlining how alt text is structured or evaluated. As such, the corpus reflects alt text as implemented in practice rather than as derived from a published institutional standard.

Two researchers independently conducted a close reading [5] of each painting and its associated alt text. We utilized a shared notetaking structure to document **1. observable formal elements** (e.g., composition, color palette, lighting, spatial arrangement, gesture, materiality) and **2. analytic observations about what the alt text prioritized or omitted**, with an example of notes taken shown in Figure 1. Researchers first generated notes independently to surface distinct interpretive perspectives, then met to compare observations and identify recurring patterns across the corpus. This approach allowed us to examine how alt text represents paintings in practice and to identify mismatches between what is visually

present in the artwork and what is conveyed through the alt text. For each painting, we viewed the image and wrote structured observational notes on the formal elements in the work, such as the subject, composition, color, lighting, etc. These notes were intended to document describable visual features that could inform an interpretation of the work while avoiding the claims of a single definitive reading. In parallel, we documented notes on each painting’s alt text, describing what it communicated and how it aligned with the visual features recorded in our observational notes. Through this process, we synthesized recurring patterns into themes that inform the preliminary insights drawn from this analysis.

## 4 Findings

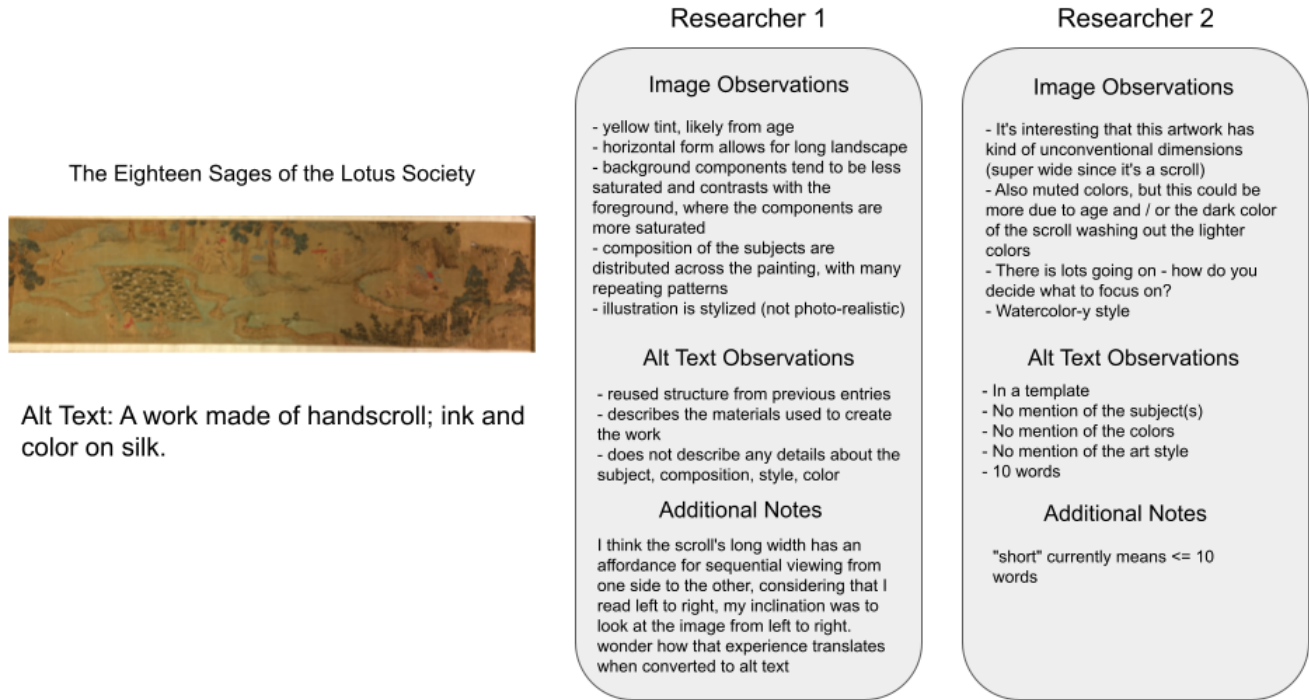
### 4.1 Inconsistencies Throughout Alt Text Practice

Across our corpus ( $n=91$ ), we found substantial inconsistency across alt texts. Rather than describing the content, most entries only provided medium and material information. We found that material-only templates ( $n=65$ ), following the pattern “a work made of *material*,” made up the majority of the dataset, followed by material with format templates ( $n=16$ ) with the pattern of “i.e., a work made of *material*; *format*.” This pattern introduces redundancy in the alt text, as material and format information is already present in other metadata fields. This templating also reduces specificity, as paintings could depict very different content despite sharing the same medium; within this corpus, the alt text “A work made of oil on canvas” appeared 28 times. The alt text was generally brief (mean 10.2 words, median 8, max 54, standard deviation 6.79). A small subset of the paintings provided depictive descriptions ( $n=10$ ) mentioning subjects, scenes, and other visual attributes regarding the content of the painting.

### 4.2 Informational Hierarchy Through Objective Formal Cues

Our analysis suggests that there is a hierarchical pattern in interpretive details, visible where formal cues observed in the paintings are prioritized in the alt text and those which were left unsupported. Details that already in metadata, such as medium and format, can help identify attributes of the artwork but contribute relatively little to the interpretive process of the work. Formal cues such as symmetry, contrast, and composition carry information that is more directly tied to how meaning is produced from the artwork while remaining objective. These cues can guide a painting’s structure without prescribing meaning.

We use these insights to construct a provisional taxonomy (Table 1) that maps out how subjective interpretations can be derived from objective, describable features of paintings. This taxonomy models a set of inferential relationships that maps observable cues to plausible interpretations, organized by **1. objective cues grounded in formal analysis** (i.e., composition, color, texture, contrast), **2. interpretive attributes that could be present from these cues** (i.e. affect, symbolism, narrative), and **3. hedging language that distinguishes interpretation from observation** (i.e., *May evoke, may suggest, typically associated*). This structure simultaneously visualizes which formal cues provide interpretive scaffolding and which



**Figure 1: Example of paired notes comparing formal visual cues observed in a painting with the content prioritized in its alt text [17].**

become more context dependence that rely on external knowledge. By mapping these relationships, this taxonomy offers a scaffold for authoring alt text for paintings while expanding interpretive access embedded within a description.

Figure 2 showcases an example using Monet's *Stack of Wheat* [15]. Though the original alt text is informative on the scene content and color palette, further of use of formal cues can support more detailed interpretive access while remaining grounded to what is observable. For instance, we observed that the wheat stack is visually isolated by wide negative space created by the low horizon line and layered background elements. Drawing on compositional principles, such spatial isolation often functions to center a subject and reduce relational context. In this case, that compositional cue may suggest solitude or loneliness. We intentionally frame this inference using hedging language ("may suggest") to distinguish interpretive possibility from a definitive claim.

## 5 Discussion

The patterns in our findings suggest that the dataset is standardized in form but inconsistent in function, and that the alt text often simply restates metadata rather than describes visual features that support interpretation. Furthermore, this templated alt text repetition can be redundant and overshadow information that supports understanding what is visually present.

When alt text defaults to these templated descriptions, it can constrain interpretive access to artworks and reinforce an access

gap between sighted viewers and BLV readers. Sighted viewers can immediately engage with the formal cues such as composition, color, and space, while BLV individuals may receive a description that offers high-level attributes without comparable support for aesthetic or affective meaning-making.

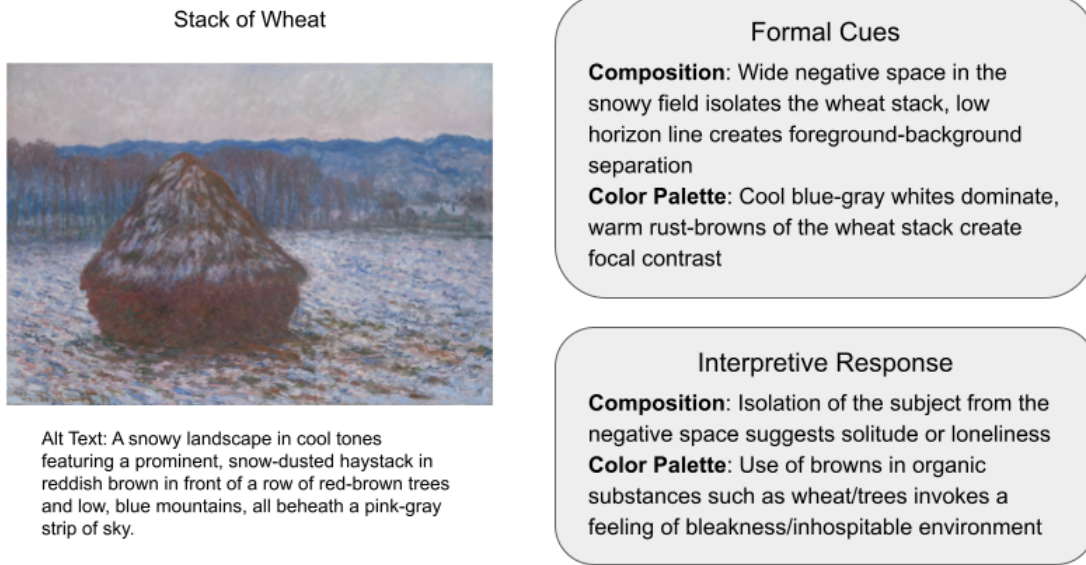
### 5.1 Taxonomy as a Design Scaffold

The provisional taxonomy offers a concrete way to address these challenges by making the observation-to-interpretation relationship explicit. By mapping objective formal cues to interpretive attributes, the taxonomy provides a structured scaffold for supporting interpretive content without presenting it as definitive fact. This can help shift alt text away from implicit, unexamined norms toward more reflective and intentional practices. This is especially relevant for institutional contexts where concerns about objectivity, authority, and neutrality can discourage interpretive description altogether. The taxonomy offers a way to navigate these concerns without collapsing interpretation into either speculative prose or purely informational catalog text.

From a technical perspective, this taxonomy can inform the design of authoring tools and automated systems. For example, an authoring interface could prompt for specific categories of formal cues, distinguish between metadata and depiction-oriented description, and support optional interpretive layers. Likewise, the taxonomy could guide or evaluate automated alt text generation systems by encouraging descriptions that move beyond generic templates

**Table 1: Taxonomy of Formal Cues and Interpretive Responses**

| Formal Cue    | Example Formal Cue           | Interpretive Response | Example Interpretation                    |
|---------------|------------------------------|-----------------------|-------------------------------------------|
| Composition   | Symmetry / depth cues        | Affect                | May suggest formality or solitude.        |
| Color & light | Lighting, contrast           | Affect                | May evoke tension.                        |
| Subject       | Skull present                | Symbolism             | Often associated with mortality.          |
| Expression    | Averted gaze, closed posture | Narrative / relation  | May read as guardedness or contemplation. |

**Figure 2: Example application of the taxonomy using Monet’s Stack of Wheat [15], showing how objective formal cues can inform interpretive responses.**

while remaining grounded in observable features. In this way, the taxonomy bridges qualitative insight and actionable system design for more equitable access to interpretive media.

## 6 Limitations and Future Work

This study draws on a single institutional corpus of 91 open-access paintings with available alt text. While this provides a view of how alt text for paintings is authored in a museum context, guidelines likely vary across institutions. We view this dataset as a useful baseline for future work. This corpus offers a concrete reference point for current painting alt text practices and can support replication across institutions, serve as a reference for longitudinal comparisons, and evaluation of future interventions by providing a benchmark for assessing coverage of formal cues and authoring approaches of interpretive language over time. Future work may expand this analysis by incorporating audio guide transcripts

and alt text from multiple institutions to examine variation across description modalities.

Building upon this baseline, we plan to conduct an interview study with BLV participants to examine how formal cues affect comprehension and aesthetic engagement compared to informational descriptions. Interviewing other stakeholders, such as museum staff, artists, and curators can offer additional perspectives on institutional constraints and authoring workflows, including how decisions on subjective interpretation are negotiated. We also plan to extend the analysis through critical discourse analysis [18] of painting alt text, examining how legitimacy is textually constructed, how objectivity-centered norms constrain interpretation, and which perspectives are prioritized or omitted. We are also interested in exploring creative writing informed alt text as a complementary mode of access for interpretive media, investigating how techniques such as imagery, metaphor, and subtext can convey affect while maintaining traceability to formal cues. We envision these future

contributions as design probes that offer multiple description styles, such as an informative baseline with an auxiliary creative interpretation, allowing participants to have agency over the style of engagement to best support their access needs.

## 7 Conclusion

In this paper, we examine how alt text for paintings can move beyond informational description to better support interpretive engagement. Through an analysis of 91 paintings from an open-access collection, we show how alt text often prioritizes templated, metadata-forward phrasing, leaving many of the formal cues that organize meaning in paintings underdescribed. We introduce a provisional taxonomy that supports a scaffolded approach for exploring how objectivity and subjectivity might be brought into relation, enabling aesthetic, narrative, and affective dimensions of artworks to be more fully represented in alt text.

We aim to contribute to ongoing conversations on accessibility as more than informational sufficiency, particularly in museum and cultural heritage contexts where description practices shape whose interpretive access is supported. By articulating a structured relationship between formal cues and interpretation, this work extends discussions of how description practices mediate aesthetic experience for BLV audiences, and how alt text can be designed to better support interpretive richness alongside accountability to what is observable.

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